Kp and Ki Test Sheets

Test it. Measure it. Write it down.

Names: _____ **Date:** _____

Our robot drives straight because the gyro corrects it many times a second. **Kp** and **Ki** are the two numbers that decide how hard it corrects. We tuned them on the bare robot. Each attachment changes the weight and balance, so the numbers might need to change. **This test finds out.** It also gives us proof for the judges: ten runs on paper beat "it seemed fine".

What are Kp and Ki?

Kp reacts to how far off straight the robot is *right now*. Too low: it drifts. Too high: it weaves left and right.

Ki reacts to how long it has been off. It fixes a steady pull to one side. Too high: it swings past straight and wobbles near the end.

What to do

1. **Set up the course** and check the setup list on page 2.
2. **Baseline:** run 5 times at the current Kp and Ki. If every run passes, skip to the confirm runs.
3. **Kp sweep:** set Ki to 0 and try five Kp values, 3 runs each. Pick the best Kp.
4. **Ki sweep:** keep that Kp and try five Ki values, 3 runs each. Pick the smallest Ki that works.
5. **Confirm:** 10 runs in a row at the chosen Kp and Ki, then the quick checks and the 30 cm drive.

The one rule

- **Change one number at a time** and repeat the exact same test. If two things change, we cannot tell which one made the difference.
- **Write down every run,** including the bad ones. A failed run is useful.

The sheets in this packet

#	Sheet	What is on it	Pages	Done
1	**Guide**	How to run the test and how to measure	2	☐
2	**No attachment (control)**	Baseline, Kp sweep, Ki sweep (sheet 1) and confirm runs (sheet 2)	3-4	☐
3	**Forklift**	Baseline, Kp sweep, Ki sweep (sheet 1) and confirm runs (sheet 2)	5-6	☐
4	**Seed bucket (lifting arm)**	Baseline, Kp sweep, Ki sweep (sheet 1) and confirm runs (sheet 2)	7-8	☐
5	**Mission 1 attachment**	Baseline, Kp sweep, Ki sweep (sheet 1) and confirm runs (sheet 2)	9-10	☐
6	**2-axis (with hook)**	Baseline, Kp sweep, Ki sweep (sheet 1) and confirm runs (sheet 2)	11-12	☐
7	**Summary**	One row per attachment, and the decision	13	☐

Hand the pages back to Chris when you are done. Time needed: about 15 minutes per attachment if the baseline passes, 45 to 60 minutes for a full tune.

Setup, measuring and what the results mean

Setup list (tick before you start)

- [ ] Same flat, clean mat or floor every time. Wipe the wheels.
- [ ] Tape a straight 100 cm line. Mark the end line.
- [ ] Start jig (two blocks or a wall edge) so every run starts in the same place.
- [ ] Sticker on the robot's front centre. Always measure to it.
- [ ] Battery charged. Write the level on the sheet.
- [ ] Attachment on with **quick pins**. Tight.
- [ ] Program is one block: **Gyro Move, 100 cm, speed 90**.
- [ ] Hold the robot still for 3 seconds after starting. Do not lift or bump it until it stops.

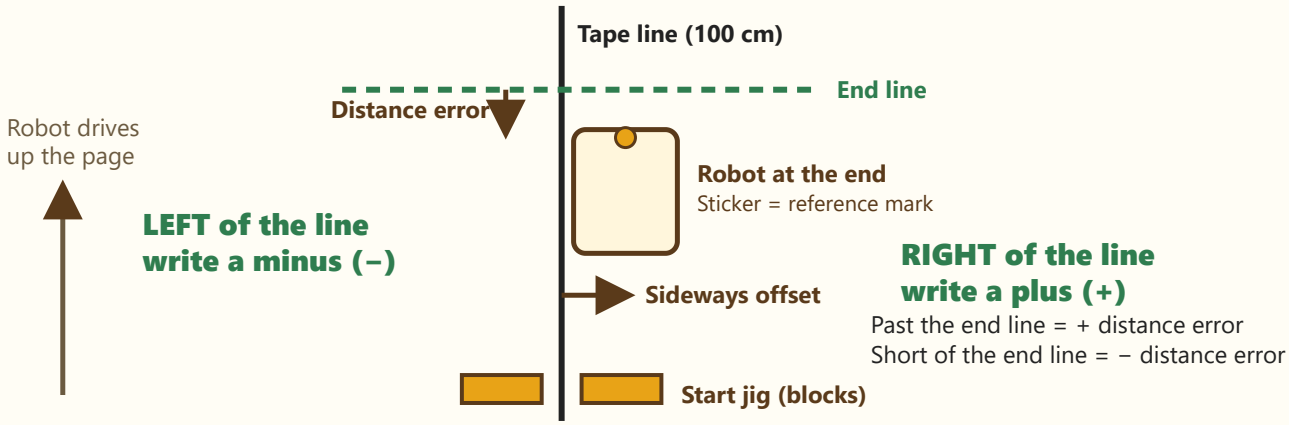
Roles (swap for each attachment)

Runner	Puts the robot on the jig and starts the program
Measurer	Reads the offset and distance error off the tape
Recorder	Writes every run on the sheet
Coder	Changes Kp or Ki in **Gyro Move** and downloads to the hub

Wobble score (watch the robot, do not guess later)

0	Dead straight
1	Tiny twitch
2	Visible weave
3	Big swings, or the attachment shakes

Where to measure



What does it mean? (Use this when a baseline run misses)

What you see	Likely cause	Change first
Ends on the same side every run, no weave	Steady pull: lopsided weight or friction	**Ki up.** Check the attachment is centred.
Weaves left and right the whole way	Kp too high	**Kp down**
Slowly curves away and Ki does not help	Kp too low	**Kp up**
Straight, then swings or wiggles at the end	Ki too high (it builds up)	**Ki down**

A pass means (suggested target, the team can change it)

- For **10 runs in a row**: every run ends within **2 cm** of the line, and the average miss (ignore + and −) is **1 cm or less**.
- Distance error within **1 cm**, wobble **1 or lower**, and **no problem runs** (attachment shifts, robot tips, seeds fall out).
- The misses are **not all on the same side.** All left means drift, not luck.

No attachment (control): sheet 1 of 2

Date: _____ **Battery:** _____ % **Runner:** _____ **Measurer:** _____

Recorder: _____ **Coder:** _____

TEST IN THIS STATE

Bare robot, no attachment. This is our reference. Every attachment is compared against it. It also catches problems that have nothing to do with attachments, like a dirty wheel or a loose ball joint.

Step 1 Baseline: 5 runs at the current Kp and Ki

Read both numbers from the code first.

Current Kp: _____ **Current Ki:** _____ If all 5 pass, skip to the confirm runs on sheet 2.

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem? (shifted, tipped, seeds fell, hook swung)
1				
2				
3				
4				
5				

Baseline: average miss _____ cm. All 5 pass? ☐ **Yes** ☐ **No**

Step 2a Kp sweep with Ki = 0

3 runs each. Write the offset in cm and the wobble score.

Kp	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
1.0								
1.5								
2.0								
2.5								
3.0								

Chosen Kp: _____ because _____

Step 2b Ki sweep with the chosen Kp

Ki 0 is the same as the chosen Kp row above: copy it.

Ki	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
0								
0.5								
1								
2								
3								

Chosen Ki: _____ smallest Ki that fixes the drift, because _____

No attachment (control): sheet 2 of 2

Step 3 **Confirm: 10 runs in a row, no changes** Do not throw out a bad run. Write why.

Final Kp: _____ **Final Ki:** _____ Test state: Bare robot, no attachment

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem?
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Average miss (ignore + / −): _____ cm. Worst run: _____ cm. Highest wobble: _____ Pass? ☐ **Yes** ☐ **No**

Step 4 **Quick check in the other states** Not needed for the control run. Go to step 5.

Step 5 **Short drive: 30 cm, 3 runs** Most missions drive a short distance. Pass = within 1 cm and no wobble.

Run	Offset (cm)	Distance error (cm)	Wobble 0 to 3
1			
2			
3			

✎ **What did this attachment do to the robot, and what did we change because of it?** ("Nothing needed to change, and here are 10 runs that prove it" is a good answer.)

Tested by: _____ **Date:** _____

Forklift: sheet 1 of 2

Date: _____ **Battery:** _____ % **Runner:** _____ **Measurer:** _____

Recorder: _____ **Coder:** _____

TEST IN THIS STATE
Forklift raised to full height, with the payload on it. The forklift is the tallest and has the highest centre of gravity. Watch for the robot rocking forward when it stops, and check the quick pins are tight before every batch of runs.

Step 1 Baseline: 5 runs at the current Kp and Ki

Read both numbers from the code first.

Current Kp: _____ **Current Ki:** _____ If all 5 pass, skip to the confirm runs on sheet 2.

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem? (shifted, tipped, seeds fell, hook swung)
1				
2				
3				
4				
5				

Baseline: average miss _____ cm. All 5 pass? ☐ **Yes** ☐ **No**

Step 2a Kp sweep with Ki = 0

3 runs each. Write the offset in cm and the wobble score.

Kp	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
1.0								
1.5								
2.0								
2.5								
3.0								

Chosen Kp: _____ because _____

Step 2b Ki sweep with the chosen Kp

Ki 0 is the same as the chosen Kp row above: copy it.

Ki	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
0								
0.5								
1								
2								
3								

Chosen Ki: _____ smallest Ki that fixes the drift, because _____

Forklift: sheet 2 of 2

Step 3 Confirm: 10 runs in a row, no changes

Do not throw out a bad run. Write why.

Final Kp: _____ **Final Ki:** _____ Test state: Forklift raised to full height, with the payload on it

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem?
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Average miss (ignore + / −): _____ cm. Worst run: _____ cm. Highest wobble: _____ Pass? ☐ **Yes** ☐ **No**

Step 4 Quick check in the other states

3 runs each at the final Kp and Ki. Write offset / wobble, like −0.5 / 1.

State	Run 1	Run 2	Run 3	Pass?
Forklift lowered (carry position), empty				☐ **Y** ☐ **N**

Step 5 Short drive: 30 cm, 3 runs

Most missions drive a short distance. Pass = within 1 cm and no wobble.

Run	Offset (cm)	Distance error (cm)	Wobble 0 to 3
1			
2			
3			

✎ **What did this attachment do to the robot, and what did we change because of it?** ("Nothing needed to change, and here are 10 runs that prove it" is a good answer.)

Tested by: _____ **Date:** _____

Seed bucket (lifting arm): sheet 1 of 2

Date: _____ **Battery:** _____ % **Runner:** _____ **Measurer:** _____

Recorder: _____ **Coder:** _____

TEST IN THIS STATE
Seed bucket loaded with the seeds in the match order. The bucket hangs out at the front, so it puts more weight on the ball joint. Watch for seeds falling out and for the robot pulling to one side.

Step 1 Baseline: 5 runs at the current Kp and Ki

Read both numbers from the code first.

Current Kp: _____ **Current Ki:** _____ If all 5 pass, skip to the confirm runs on sheet 2.

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem? (shifted, tipped, seeds fell, hook swung)
1				
2				
3				
4				
5				

Baseline: average miss _____ cm. All 5 pass? ☐ **Yes** ☐ **No**

Step 2a Kp sweep with Ki = 0

3 runs each. Write the offset in cm and the wobble score.

Kp	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
1.0								
1.5								
2.0								
2.5								
3.0								

Chosen Kp: _____ because _____

Step 2b Ki sweep with the chosen Kp

Ki 0 is the same as the chosen Kp row above: copy it.

Ki	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
0								
0.5								
1								
2								
3								

Chosen Ki: _____ smallest Ki that fixes the drift, because _____

Seed bucket (lifting arm): sheet 2 of 2

Step 3 **Confirm: 10 runs in a row, no changes** Do not throw out a bad run. Write why.

Final Kp: _____ **Final Ki:** _____ Test state: Seed bucket loaded with the seeds in the match order

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem?
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Average miss (ignore + / −): _____ cm. Worst run: _____ cm. Highest wobble: _____ Pass? ☐ **Yes** ☐ **No**

Step 4 **Quick check in the other states** 3 runs each at the final Kp and Ki. Write offset / wobble, like −0.5 / 1.

State	Run 1	Run 2	Run 3	Pass?
Bucket empty, arm lowered				☐ **Y** ☐ **N**
Bucket empty, arm raised				☐ **Y** ☐ **N**

Step 5 **Short drive: 30 cm, 3 runs** Most missions drive a short distance. Pass = within 1 cm and no wobble.

Run	Offset (cm)	Distance error (cm)	Wobble 0 to 3
1			
2			
3			

✎ **What did this attachment do to the robot, and what did we change because of it?** ("Nothing needed to change, and here are 10 runs that prove it" is a good answer.)

Tested by: _____ **Date:** _____

Mission 1 attachment: sheet 1 of 2

Date: _____ **Battery:** _____ % **Runner:** _____ **Measurer:** _____

Recorder: _____ **Coder:** _____

Which build is it? _____

TEST IN THIS STATE
As carried on the way to Mission 1. Write which build this is (Giraffes #3388, Platinum Pizzeria or Team TurtleBots) so the result goes with the right attachment.

Step 1 Baseline: 5 runs at the current Kp and Ki

Read both numbers from the code first.

Current Kp: _____ **Current Ki:** _____ If all 5 pass, skip to the confirm runs on sheet 2.

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem? (shifted, tipped, seeds fell, hook swung)
1				
2				
3				
4				
5				

Baseline: average miss _____ cm. All 5 pass? ☐ **Yes** ☐ **No**

Step 2a Kp sweep with Ki = 0

3 runs each. Write the offset in cm and the wobble score.

Kp	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
1.0								
1.5								
2.0								
2.5								
3.0								

Chosen Kp: _____ because _____

Step 2b Ki sweep with the chosen Kp

Ki 0 is the same as the chosen Kp row above: copy it.

Ki	Run 1		Run 2		Run 3		Average miss (ignore + / −)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
0								
0.5								
1								
2								
3								

Chosen Ki: _____ smallest Ki that fixes the drift, because _____

Mission 1 attachment: sheet 2 of 2

Step 3 Confirm: 10 runs in a row, no changes

Do not throw out a bad run. Write why.

Final Kp: _____ **Final Ki:** _____ Test state: As carried on the way to Mission 1

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem?
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Average miss (ignore + / −): _____ cm. Worst run: _____ cm. Highest wobble: _____ Pass? ☐ **Yes** ☐ **No**

Step 4 Quick check in the other states

3 runs each at the final Kp and Ki. Write offset / wobble, like −0.5 / 1.

State	Run 1	Run 2	Run 3	Pass?
Any other arm position it uses: _____				☐ **Y** ☐ **N**

Step 5 Short drive: 30 cm, 3 runs

Most missions drive a short distance. Pass = within 1 cm and no wobble.

Run	Offset (cm)	Distance error (cm)	Wobble 0 to 3
1			
2			
3			

✎ **What did this attachment do to the robot, and what did we change because of it?** ("Nothing needed to change, and here are 10 runs that prove it" is a good answer.)

Tested by: _____ **Date:** _____

2-axis (with hook): sheet 1 of 2

Date: _____ **Battery:** _____ % **Runner:** _____ **Measurer:** _____

Recorder: _____ **Coder:** _____

TEST IN THIS STATE
Both axes fully extended. The hook sticks out in front. Watch for it swinging, and for the weight shifting when the axes are extended.

Step 1 Baseline: 5 runs at the current Kp and Ki

Read both numbers from the code first.

Current Kp: _____ **Current Ki:** _____ If all 5 pass, skip to the confirm runs on sheet 2.

Run	Offset (cm) right +, left –	Distance error (cm)	Wobble (0 to 3)	Problem? (shifted, tipped, seeds fell, hook swung)
1				
2				
3				
4				
5				

Baseline: average miss _____ cm. All 5 pass? ☐ **Yes** ☐ **No**

Step 2a Kp sweep with Ki = 0

3 runs each. Write the offset in cm and the wobble score.

Kp	Run 1		Run 2		Run 3		Average miss (ignore + / –)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
1.0								
1.5								
2.0								
2.5								
3.0								

Chosen Kp: _____ because _____

Step 2b Ki sweep with the chosen Kp

Ki 0 is the same as the chosen Kp row above: copy it.

Ki	Run 1		Run 2		Run 3		Average miss (ignore + / –)	Notes
	Offset	Wob.	Offset	Wob.	Offset	Wob.		
0								
0.5								
1								
2								
3								

Chosen Ki: _____ smallest Ki that fixes the drift, because _____

2-axis (with hook): sheet 2 of 2

Step 3 Confirm: 10 runs in a row, no changes

Do not throw out a bad run. Write why.

Final Kp: _____ **Final Ki:** _____ Test state: Both axes fully extended

Run	Offset (cm) right +, left −	Distance error (cm)	Wobble (0 to 3)	Problem?
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Average miss (ignore + / −): _____ cm. Worst run: _____ cm. Highest wobble: _____ Pass? ☐ **Yes** ☐ **No**

Step 4 Quick check in the other states

3 runs each at the final Kp and Ki. Write offset / wobble, like −0.5 / 1.

State	Run 1	Run 2	Run 3	Pass?
Both axes retracted				☐ **Y** ☐ **N**

Step 5 Short drive: 30 cm, 3 runs

Most missions drive a short distance. Pass = within 1 cm and no wobble.

Run	Offset (cm)	Distance error (cm)	Wobble 0 to 3
1			
2			
3			

✎ **What did this attachment do to the robot, and what did we change because of it?** ("Nothing needed to change, and here are 10 runs that prove it" is a good answer.)

Tested by: _____ **Date:** _____

Summary: the whole picture

Copy the results from each attachment's sheet 2 into one table. Judges can read this page in 30 seconds.

Attachment	Baseline Kp / Ki	Baseline avg miss (cm)	Final Kp	Final Ki	Final avg miss, 10 runs (cm)	Worst wobble	Pass? Y / N
No attachment (control)							
Forklift							
Seed bucket (lifting arm)							
Mission 1 attachment							
2-axis (with hook)							

Do the attachments need different numbers?

☐ **One shared set.** The same Kp and Ki pass on every attachment. (Try this first. If the best numbers are within about 0.5 of each other, use the set with the best average.)

Kp _____ **Ki** _____

☐ **Different numbers per attachment.** We will add Kp and Ki as inputs to **Gyro Move** and pass the right pair from each mission program. Keep a copy of the working block first.

✎ **What did we learn, and what would we do differently?** (Which attachment was hardest on the robot? Did anything surprise us?)

✎ **Still to do** (a test that failed, a part to fix, a number to confirm with Jordan)

Completed by: _____ **Date:** _____

Where do these results go?

Chris will copy the final Kp and Ki into the tuning record and add one short entry per attachment to the robot design log. Judges look for repeated testing, a change made for a reason, and a result.